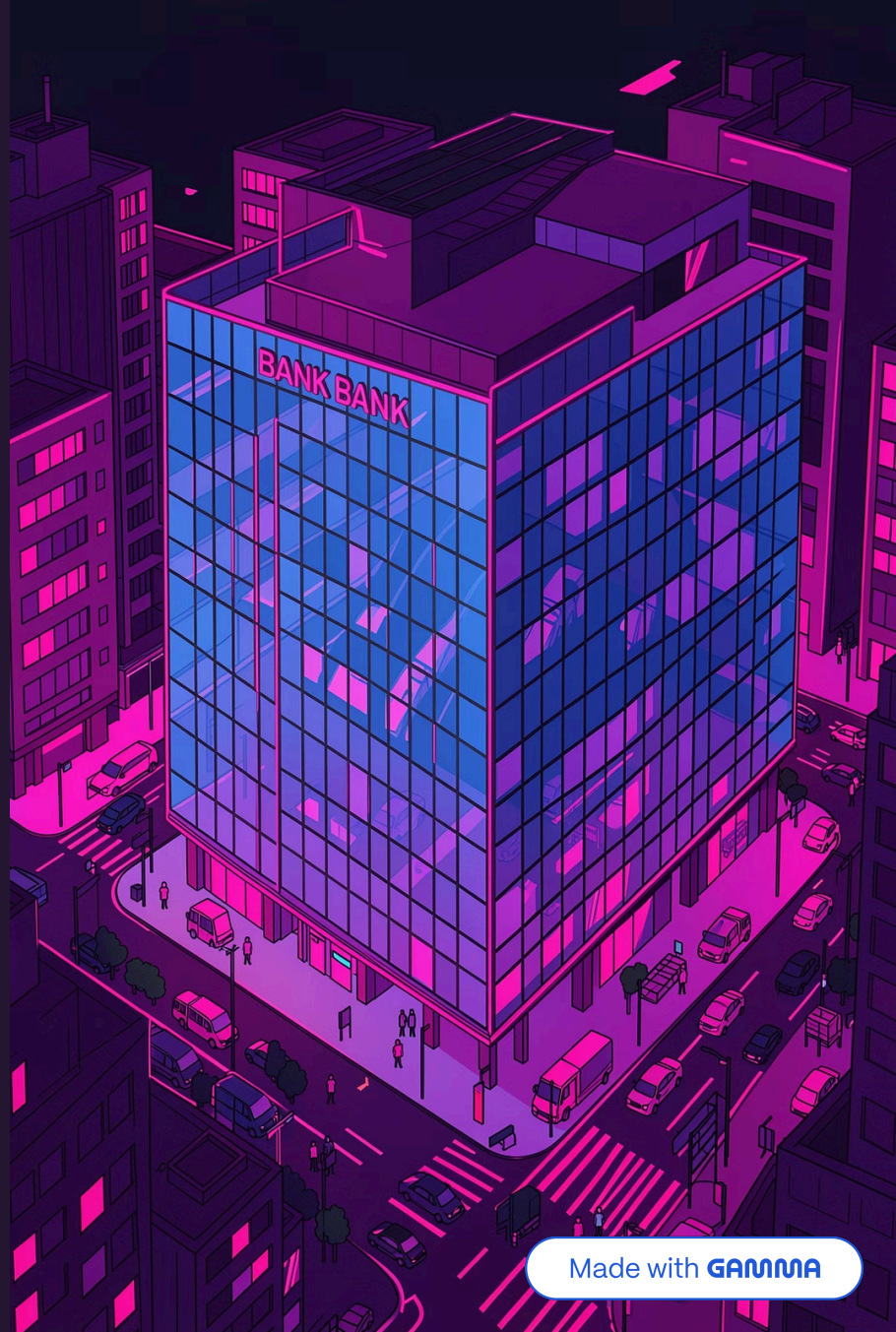
Banking Client Analytics Report

Exploratory Data Analysis, SQL Query Documentation & Dashboard Reporting

FINANCE DATABASE | 2,881 CLIENT RECORDS | 28 VARIABLES | APRIL 2026



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Executive Summary

2,881
Total Clients

22
Investment Advisors

£171K
Avg Est. Income

£670K
Avg Bank Deposits

£867K
Avg Business Lending

Top Banking Type

Private Bank — 1,286 clients (44.64%)

Largest Nationality

European — 43.63% of client base

Gender Split

Female 50.47% / Male 49.53% — near-equal

Highest Avg Balance

Business Lending — £866,569 per client

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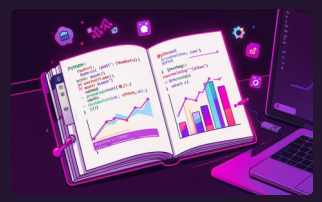
CHAPTER 1

Project Overview & Technology Stack



Python EDA

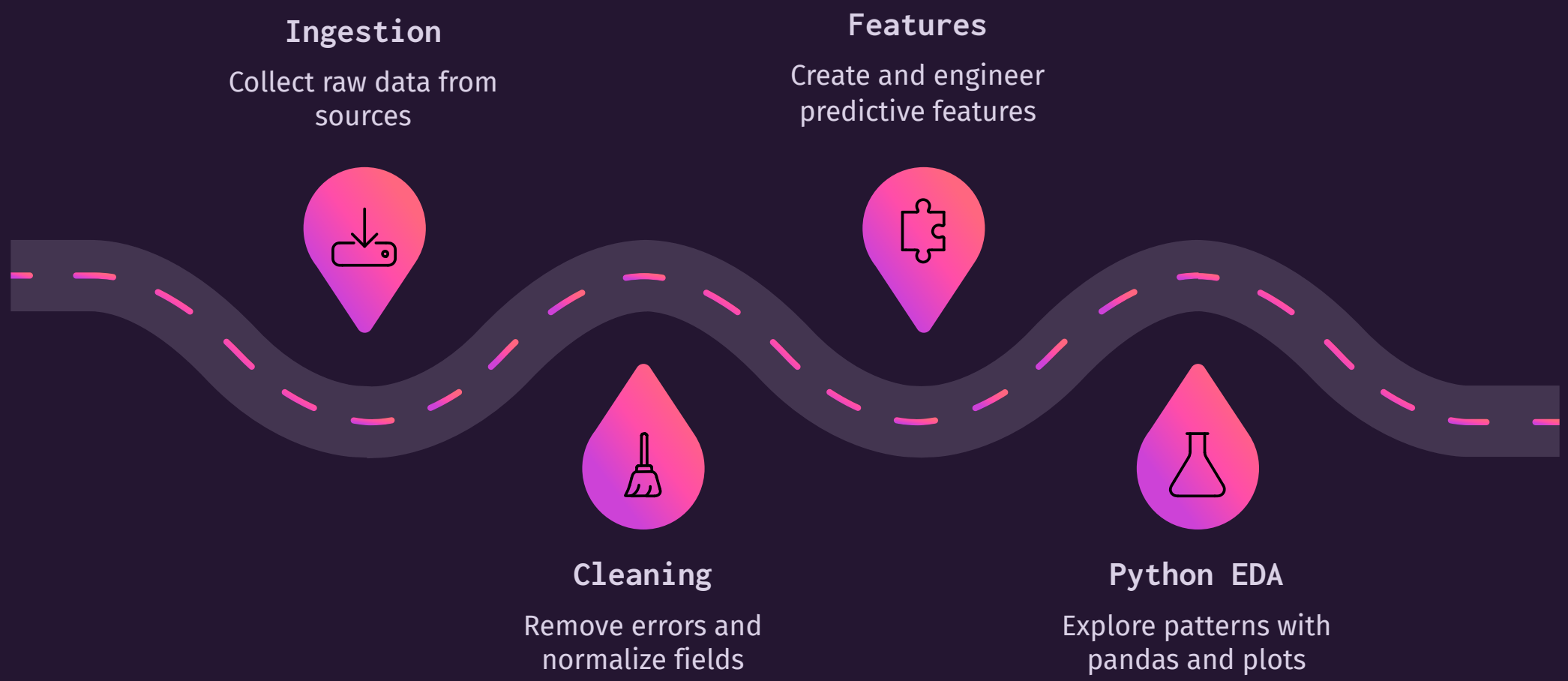
pandas, numpy, seaborn, matplotlib — data cleaning, feature engineering, visualisation

MySQL Analysis

115 unique queries across 15 sections — aggregations, CTEs, window functions, business scenarios

Power BI Dashboard

Interactive executive reporting — KPI cards, slicers, demographic charts, advisor league table



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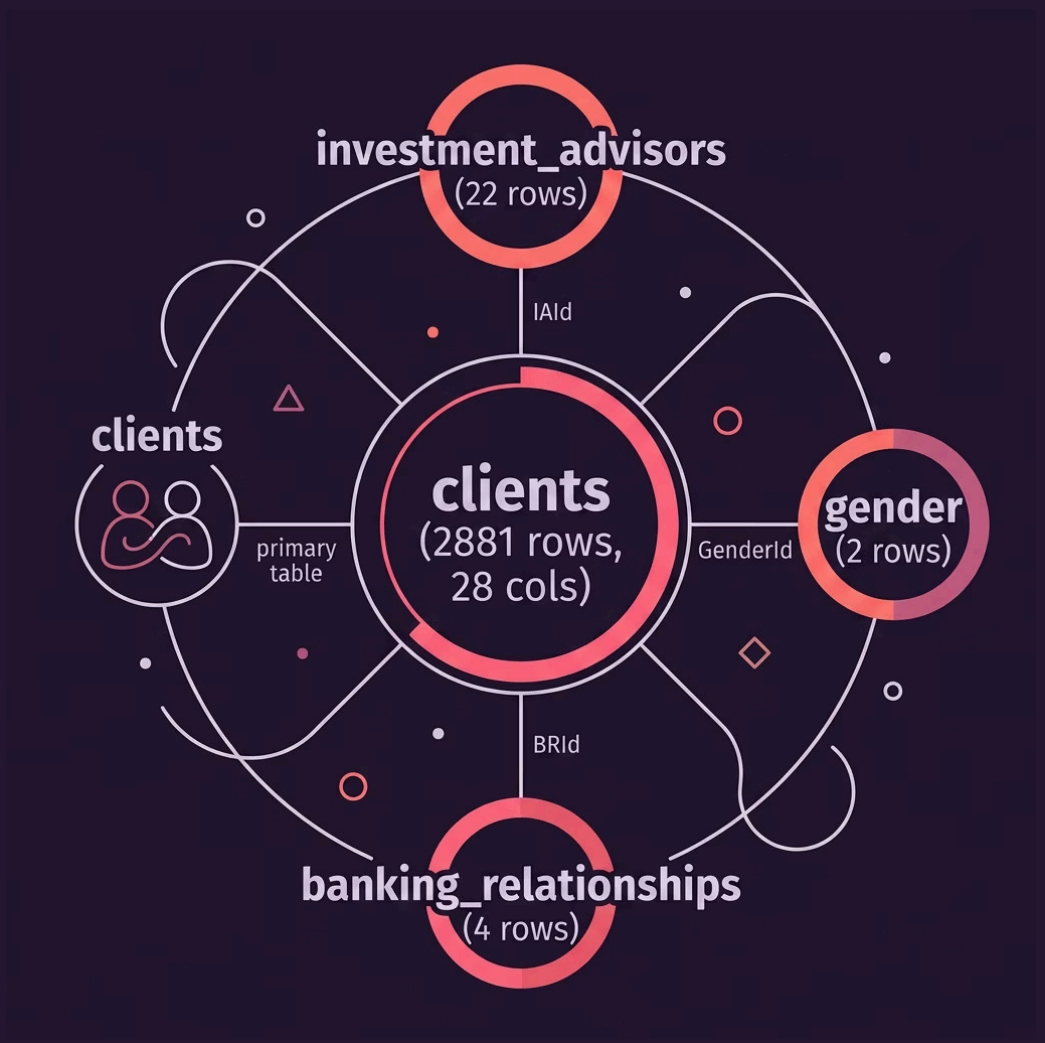
CHAPTER 2

Dataset Description



Primary Table: clients.csv

2,881 rows × 28 columns — client demographics, financial account balances, and 3 derived/engineered columns.

Derived Features

- **Income Band** — Low / Medium / High via pd.cut()
- **debt_to_income** — avg 4.7467, max 41.2155
- **savings_ratio** — avg 1.8271, max 20.1919

Dimension Tables

- gender.csv — 2 rows
- investment_advisors.csv — 22 rows
- banking_relationships.csv — 4 rows

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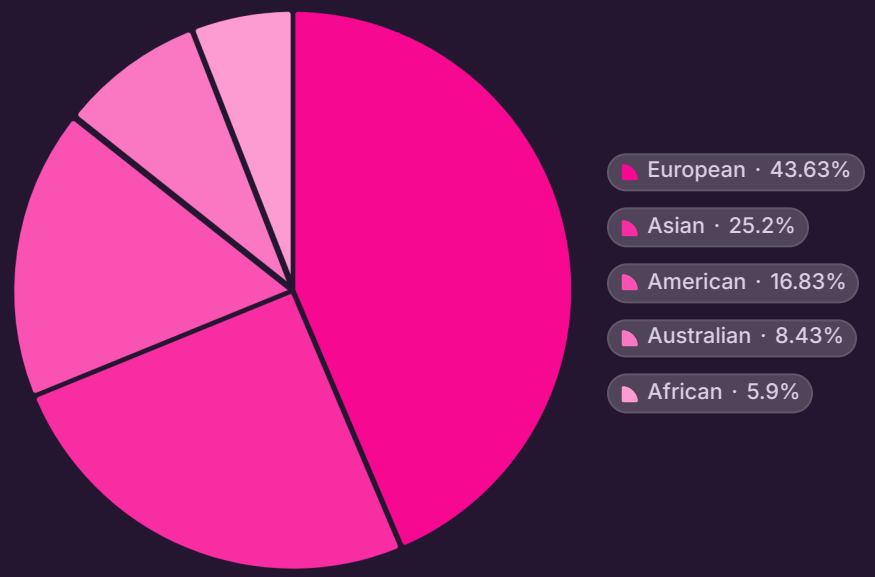
CHAPTER 3

Client Demographics



Key Demographics

- **Age:** Range 17–85, average 50.98
- **Loyalty:** Jade 44.32% → Platinum 10.45%
- **Fee Structure:** High 48.98% | Mid 32.07% | Low 18.95%
- **Income Band:** Medium 50.19% | Low 34.40% | High 15.41%
- **Risk:** Levels 1–2 = 68.48% of clients
- **Credit Cards:** 1 card = 63.94% of clients

ⓘ 195 unique occupations — no single occupation exceeds 0.9% of the client base.

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Financial Behaviour & Key Metrics

Business Lending

Highest avg balance: **£866,569** — exceeds deposits (£670K) and loans (£593K)

Deposits vs Loans

Avg deposits **£670,232** exceed avg loans **£593,265** — positive net position

Leverage

Avg debt-to-income **4.75×**; max 41.2× — some clients carry extreme leverage

Savings Ratio

Avg **1.83×** annual income; max 20.19× — large investment/retirement accounts

All 9 financial variables exhibit positive (right) skew — a small number of ultra-high-balance clients pull averages above the median.

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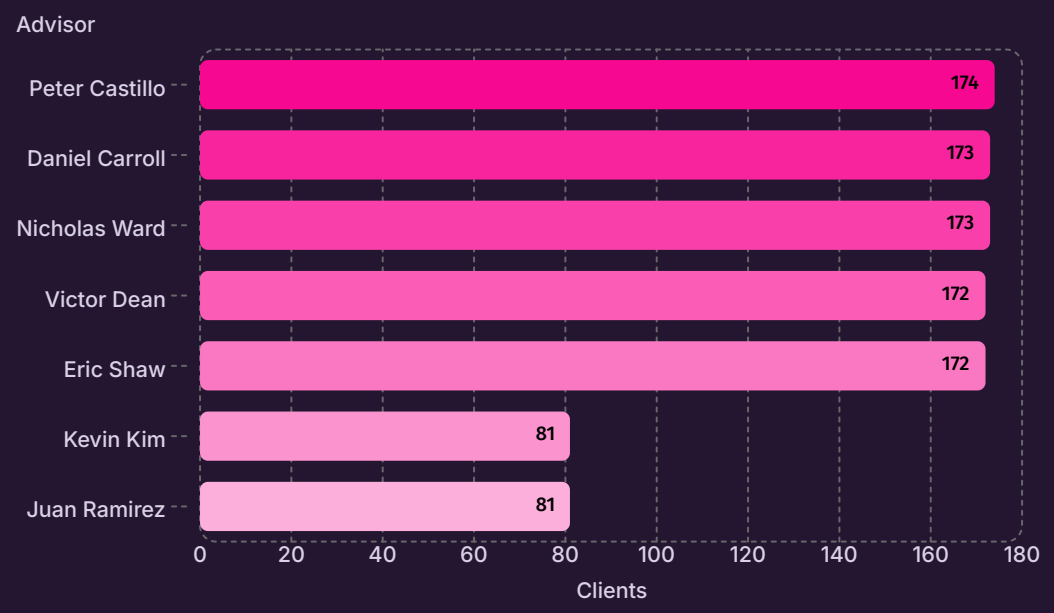
Investment Advisor Performance



Portfolio Summary

- **22 active advisors** — all have clients assigned
- Average load: **130.95 clients**
- Std deviation: **44.29** — meaningful inequality
- Load range: **81–174** (2.15× ratio)
- Top 5 advisors manage **29.99%** of all clients
- All 22 advisors serve **both genders**

⚠ The 2.15× load imbalance warrants rebalancing review — top advisors carry more than double the lightest portfolios.

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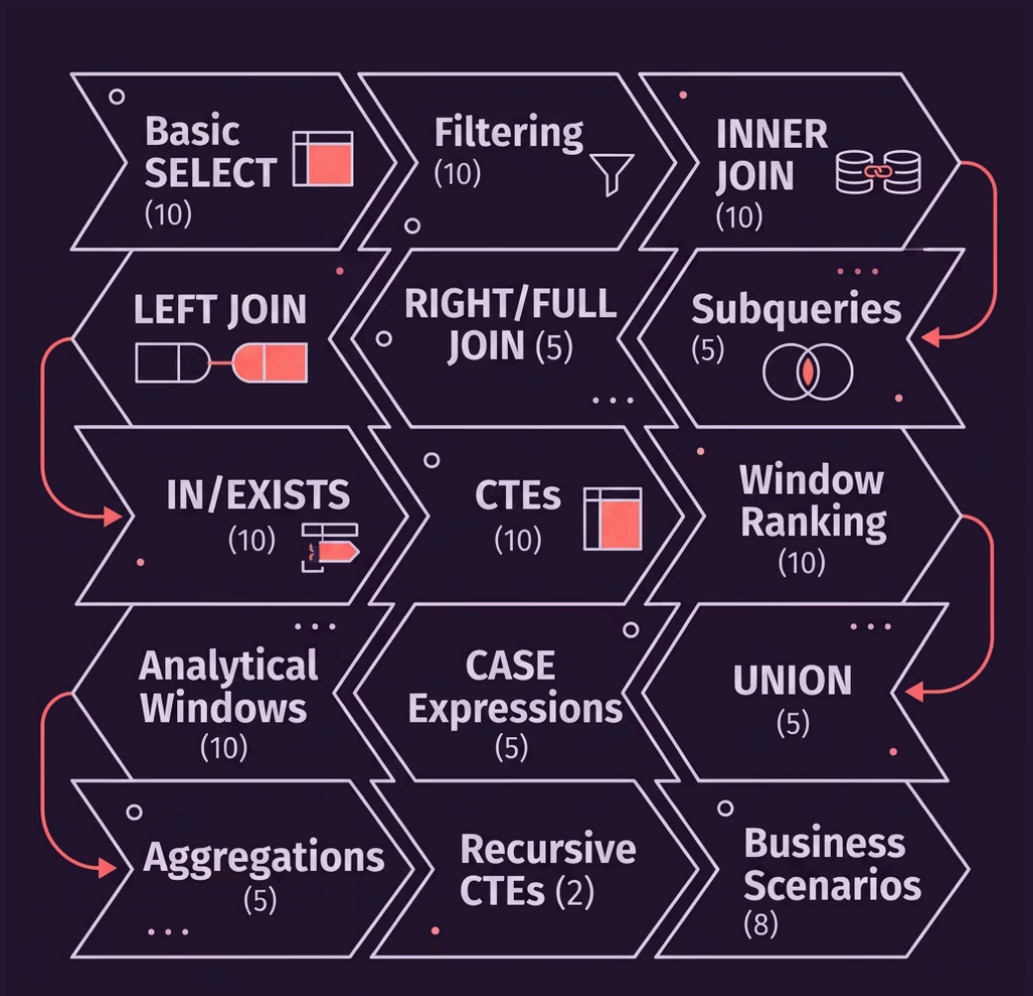
MySQL Analysis: 115 Queries, 15 Sections



Notable Query Patterns

Query 46 – Correlated Subquery

3-level nesting: clients whose advisor exceeds average load

Query 75 – 4-CTE Chain

Cumulative client distribution with progressive window aggregation

Query 115 – Gap Analysis

CROSS JOIN: 22 × 2 × 4 = 176 combinations; surfaces uncovered segments

Query 120 – Exec Dashboard

5 CTEs combined via UNION ALL into a single KPI result set

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Power BI Dashboard & Data Quality

Dashboard Visuals

KPI Cards

Total clients, advisors, avg income, deposits, business lending

Demographic Charts

Nationality, loyalty, fee structure, top occupations

Slicers

Nationality, gender, income band, banking relationship, loyalty, risk

Correlation Charts

Income vs deposits; loans vs business lending scatter plots

Data Quality Results

Check	Result
Missing values	Zero across all 28 columns
Duplicate records	Zero in 2,881 rows
Invalid FK references	None — all resolve correctly
Advisors with no clients	Zero — all 22 active
Unused banking types	None — all 4 BRIds used

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CHAPTER 8

Recommendations

1

Client Retention & Loyalty Upgrade

Target Medium-income Jade clients (44.32%) on High/Mid fees for upgrade. Maintain dedicated capacity for 301 Platinum clients. Silver tier (739) is the clearest Gold-upgrade opportunity.

2

Advisor Capacity Rebalancing

Redistribute 20–25 clients from top advisors (174) to bottom advisors (81). Schedule SQL Queries 116–117 as monthly automated concentration-risk reports.

3

Product Strategy

Commercial + Institutional = only 33.08% combined vs Private Bank at 44.64%. Avg debt-to-income of 4.75× signals opportunity for loan consolidation and refinancing products.

4

Analytics Improvements

Add cohort analysis using Joined_Bank date. Complete missing SQL queries 51–55. Connect Power BI directly to MySQL for live refresh. Integrate transaction-level data for churn prediction.

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